

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 12

Question 1 - MCQ · PG-20 · LR-1

Partition values divide a distribution into

Options

- A. a required number of equal parts
- B. unequal classes only
- C. two parts only
- D. random groups

Correct answer: a required number of equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives a required number of equal parts.

Question 2 - MCQ · PG-20 · LR-2

Quartiles divide a distribution into

Options

- A. 4 equal parts
- B. 10 equal parts
- C. 100 equal parts
- D. 2 equal parts

Correct answer: 4 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 4 equal parts.

Question 3 - MCQ · PG-20 · LR-3

Deciles divide a distribution into

Options

- A. 10 equal parts
- B. 4 equal parts
- C. 100 equal parts
- D. 9 equal parts

Correct answer: 10 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 10 equal parts.

Question 4 - MCQ · PG-20 · LR-4

Percentiles divide a distribution into

Options

- A. 100 equal parts
- B. 10 equal parts
- C. 99 equal parts
- D. 4 equal parts

Correct answer: 100 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 100 equal parts.

Question 5 - MCQ · PG-20 · LR-5

For raw data, the position of Q_i is

Options

- A. $(i (n + 1))/(4)$
- B. $(i N)/(4) - \text{c.f.}$
- C. $(i (n + 1))/(10)$
- D. $(i (n + 1))/(100)$

Correct answer: $(i (n + 1))/(4)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i (n + 1))/(4)$.

Question 6 - MCQ · PG-20 · LR-6

For raw data, the position of D_i is

Options

- A. $(i (n + 1))/(10)$
- B. $(i (n + 1))/(4)$
- C. $(i N)/(10) - \text{c.f.}$
- D. $(i (n + 1))/(100)$

Correct answer: $(i (n + 1))/(10)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i (n + 1))/(10)$.

Question 7 - MCQ · PG-20 · LR-7

For raw data, the position of P_i is

Options

- A. $(i + 1)/(100)$
- B. $(i + 1)/(10)$
- C. $(i + 1)/(4)$
- D. $(i N)/(100) - c.f.$

Correct answer: $(i + 1)/(100)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i + 1)/(100)$.

Question 8 - MCQ · PG-20 · LR-8

For grouped data, Q_i uses the class containing

Options

- A. $(i N)/(4)$
- B. $(i N)/(10)$
- C. $(i N)/(100)$
- D. $(N)/(i)$

Correct answer: $(i N)/(4)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(4)$.

Question 9 - MCQ · PG-20 · LR-9

Approximate partition values can be determined using an

Options

- A. ogive
- B. ordinary bar chart
- C. pie chart
- D. pictograph

Correct answer: ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives ogive.

Question 10 - MCQ · PG-20 · LR-10

Which identity is correct?

Options

A. Median = $Q_2 = D_5 = P_{50}$

B. Median = $Q_1 = D_5$

C. Median = $Q_3 = P_{75}$

D. Median = $D_1 = P_{10}$

Correct answer: Median = $Q_2 = D_5 = P_{50}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Median = $Q_2 = D_5 = P_{50}$.

Question 11 - MCQ · PG-20 · LR-11

The correct quartile order is

Options

A. $Q_1 < Q_2 < Q_3$

B. $Q_3 < Q_2 < Q_1$

C. $Q_1 = Q_2 = Q_3$

D. $Q_2 < Q_1 < Q_3$

Correct answer: $Q_1 < Q_2 < Q_3$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_1 < Q_2 < Q_3$.

Question 12 - MCQ · PG-20 · LR-12

The correct decile order begins

Options

A. $D_1 < D_2 < D_3 < \dots < D_9$

B. $D_9 < D_8 < \dots < D_1$

C. $D_1 = D_2 = \dots = D_9$

D. $D_1 < Q_1 < P_1$ only

Correct answer: $D_1 < D_2 < D_3 < \dots < D_9$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $D_1 < D_2 < D_3 < \dots < D_9$.